

# *Exploratory Data Analysis*

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## *Introduction*

This Rmarkdown file is part of the **CommuteCA** package. This package was created in conjunction with the office of the *Research Data Center* at *McMaster University*, the *Sherman Centre for Digital Scholarship* and the *Mobilizing Justice*<sup>1</sup>.

The **CommuteCA** R package was created to develop standardized methods for transport analysis in research, particularly for analysis using the *2021 Census of Population* from Statistics Canada. We focused our efforts on the *Commuting Reference Guide*, which provides valuable variables and information on commuting for the Canadian population aged 15 and older living in private households.

## *Section objectives*

This R markdown aims to provide a brief introduction to exploratory data analysis (EDA). This notebook is an updated and reduced version of the computational notebooks available in the *edashop* package. The *edashop* package is an open educational resource to teach a workshop on EDA using R and computational notebooks, created and maintained by Dr. Antonio Paez, co-author of the *CommuteCA* package. For those interested in learning EDA in more deep, we suggest installing and studying the *edashop* package.

In this first R markdown, we will learn about:

- Descriptive statistics
- Visualization techniques

## *Preliminaries*

Clear the workspace from *all* objects:

```
rm(list = ls())
```

Load packages. Remember, packages are units of shareable code that augment the functionality of base R. For this session, the following packages are used:

```
library(CommuteCA) # To access data sets
library(corr) # Correlations in R
library(dplyr) # A Grammar of Data Manipulation
```

<sup>1</sup> The Mobilizing Justice project is a multidisciplinary and multi-sector collaboration with the objective of understand and address transportation poverty in Canada and to improve the well-being of Canadians at risk of transport poverty. The Social Sciences and Humanities Research Council (SSRHC) has provided funding for the project, which was created by an unprecedented alliance of academics from various Canadian provinces and institutions, transportation firms, and nonprofit organizations

```
library(janitor) # Simple Tools for Examining and Cleaning Dirty Data
library(ggplot2) # Create Elegant Data Visualizations Using the Grammar of Graphics
library(ggridges) # Ridgeline Plots in 'ggplot2'
library(readr) # Read csv files
library(skimr) # Summary statistics about variables in data frames
library(here) # enable easy file referencing in project-oriented workflows
library(DescTools) # Tools for Descriptive Statistics
library(poliscidata) # Weights in R
library(Hmisc) # Weights in R
library(tidyr) # Pivoting tables
library(weights) # Weights in R
```

We will also load the following data frames for this session. The dataset used in this demonstration is test data produced to replicate the variables available in the original Census of Population. The test data contains 200,000 rows and 17 columns. As in the original census data, each row refers to a respondent and each column refers to a variable<sup>2</sup>.

The creation of test data was necessary because the surveys provided by Statistics Canada are confidential and cannot be accessed outside of a Research Data Center.

If you want to work with the original Census dataset, the process for performing exploratory data analysis will be the same as for the test data, except that you will have to update the address of the file in the chunk<sup>3</sup> called *load-census-data*.

For this R markdown, we'll use the following variables<sup>4</sup>:

| Variable | Description                                                                                        |
|----------|----------------------------------------------------------------------------------------------------|
| PRCDDA   | Refers to the dissemination area (DA) of current residence.                                        |
| Pr       | Refers to the province or territory of current residence.                                          |
| CMA      | Census metropolitan area or census agglomeration of current residence.                             |
| PCD      | Census division of current residence.                                                              |
| CompW1   | Weight for the households and dwellings universes.                                                 |
| LBR_FORC | This variable refers to whether a person was employed, unemployed or not in the labour force.      |
| CfInc_AT | Total income of census family (sum of the total incomes of all members of that family), after tax. |
| CFCNT    | Census family size (numbers of persons).                                                           |

<sup>2</sup> You can check out more information about the Census on the Dictionary website.

<sup>3</sup> A code chunk is an executable part of the R code.

<sup>4</sup> The explanation of each variable can be found in the *2021 Census of Population's website*.

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|                     |                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| CF_PnCF_NumEarnings | Number of earners in census family.                                                                                             |
| PWDA                | Place of work dissemination area.                                                                                               |
| PWPR                | Place of work province.                                                                                                         |
| PWCMA               | Census metropolitan area or census agglomeration of place of work.                                                              |
| PWCD                | Place of work census division.                                                                                                  |
| PWDUR               | Commuting duration, it refers to the length of time, in minutes, usually required by a person to travel to their place of work. |
| PWDist              | Distance (straight-line) from home to work.                                                                                     |
| PwMode              | Main mode of commuting' refers to the main mode of transportation a person uses to travel to their place of work.               |

---

We will use the function `read_csv()` to read the sheets in the format .csv . This is one of the format used by Research Data Centres to store their surveys.<sup>5</sup>

The next chunk assigns the file address of the survey to the `census_address` variable. As mentioned before, for this demonstration we will use the test version of the survey. If you want to use the original survey, please remember to *update* the address in the chunk below. You may also need to update the file address if you opened this R markdown in another project or if the test data is stored in a different folder that does not follow the pattern of the `COMMUCECA21` package.

<sup>5</sup> Other formats also used are: SAS, Stata and SPSS. For this case, use the {foreign} package.

```
# census_address <- paste0(here::here(), "census-address.csv")
# census <- read_csv(census_address)

data(census_test_data)
census <- census_test_data
```

**NOTE:** If the code above did not run correctly, you probably are experiencing a file address error. Try to identify the correct address and update the chunk named `census-file-address` to continue.

### *Exploratory Data Analysis (EDA)*

Exploratory Data Analysis (EDA) involves examining data to uncover its inherent features without relying on preconceived assumptions. John W. Tukey described EDA as detective work, where the goal is to discover key evidence and patterns before testing hypotheses, similar to the confirmation of evidence in a trial.

Effective EDA focuses on:

1. Simplifying descriptions for better cognitive processing.

## 2. Exploring deeper aspects of the data to enhance understanding.

The primary tools for EDA are descriptive statistics and visualization techniques, with a particular focus on using appropriate descriptors for different types of data.

### *Descriptive statistics*

In the previous session we used the function `summary()` from base R to obtain quick summaries of data. These summaries already provided some key information about the data, including descriptive statistics. For example, for our census table we have<sup>6</sup>:

```
census <- census %>%
  select("Frame_ID", "PRCDDA", "Pr", "CMA", "PCD", "CompW1", "LBR_FORC", "CfInc_AT", "CFCNT", "CF_PnCF_NumEarners", "PWDA", "PWPR", "PWCD", "PWCMA", "PWDUR", "PWDist")

summary(census)
```

|    |                |                    |                  |                  |
|----|----------------|--------------------|------------------|------------------|
| ## | Frame_ID       | PRCDDA             | Pr               | CMA              |
| ## | Min. : 1       | Min. :10010165     | Min. :10.00      | Length:200000    |
| ## | 1st Qu.: 50001 | 1st Qu.:35061322   | 1st Qu.:35.00    | Class :character |
| ## | Median :100001 | Median :35204479   | Median :35.00    | Mode :character  |
| ## | Mean :100001   | Mean :37275793     | Mean :37.02      |                  |
| ## | 3rd Qu.:150000 | 3rd Qu.:47150274   | 3rd Qu.:47.00    |                  |
| ## | Max. :200000   | Max. :62080028     | Max. :62.00      |                  |
| ## | PCD            | CompW1             | LBR_FORC         | CfInc_AT         |
| ## | Min. :1001     | Min. : 1.0         | Min. : -3.000    | Min. : 282       |
| ## | 1st Qu.:3506   | 1st Qu.:136.0      | 1st Qu.: 1.000   | 1st Qu.: 35665   |
| ## | Median :3520   | Median :271.0      | Median : 1.000   | Median : 59022   |
| ## | Mean :3727     | Mean :270.6        | Mean : 1.152     | Mean : 64667     |
| ## | 3rd Qu.:4715   | 3rd Qu.:406.0      | 3rd Qu.: 2.000   | 3rd Qu.: 87924   |
| ## | Max. :6208     | Max. :540.0        | Max. : 3.000     | Max. :246147     |
| ## | CFCNT          | CF_PnCF_NumEarners | PWDA             | PWPR             |
| ## | Min. :3.000    | Min. :1.0          | Min. : -3        | Min. : -3.00     |
| ## | 1st Qu.:3.000  | 1st Qu.:2.0        | 1st Qu.: -3      | 1st Qu.: -3.00   |
| ## | Median :4.000  | Median :2.0        | Median :24940381 | Median :24.00    |
| ## | Mean :4.001    | Mean :2.5          | Mean :24202796   | Mean :22.98      |
| ## | 3rd Qu.:5.000  | 3rd Qu.:3.0        | 3rd Qu.:35370624 | 3rd Qu.:35.00    |
| ## | Max. :5.000    | Max. :4.0          | Max. :62080015   | Max. :62.00      |
| ## | PWCD           | PWCMA              | PWDUR            | PWDist           |
| ## | Min. : -3      | Length:200000      | Min. : -3.00     | Min. : -3        |
| ## | 1st Qu.: -3    | Class :character   | 1st Qu.: -3.00   | 1st Qu.: -3      |
| ## | Median :2494   | Mode :character    | Median : 61.00   | Median :23132    |
| ## | Mean :2419     |                    | Mean : 57.88     | Mean :21311      |

<sup>6</sup> We use a variable named `selected_columns` to selected some columns in the census survey. This is needed because, if you are using the original census survey, the full table summary can be very slow and will require a lot of RAM from your computer.

```
## 3rd Qu.:3537          3rd Qu.:102.00  3rd Qu.:38796
## Max.      :6208          Max.      :178.00  Max.      :50000
##      PwMode
## Min.      :-3.000
## 1st Qu.   :-3.000
## Median   : 1.000
## Mean      : 1.071
## 3rd Qu.   : 3.000
## Max.      :10.000
```

The summary already gives us a important piece of information: in many variables, the value `-3` appears as the minimum value. However, this value does not means the minimum value of this variables is `-3` (how could we explain a distance value equal to `-3`?). The `-3`, in this case, means that the variable does not apply to that respondent, and is a pattern used by Statistics Canada to signalize a not applicable situation to that respondent.

For instance, one person who does not have a job will present a `-3` for the variable related to place of work in the province, showing a situation not applicable to that respondent.

The summary already shows that the variables are not classified correctly. The variable `LBR_FORC`, for instance, is a nominal categorical variable, but it is presented as a numeric variable.

This summary quickly shows why it is important to apply EDA techniques before performing more sophisticated analyses: the data can be organized in a way that will require pre-processing and manipulation; or having patterns and formats established by the creators that are important to know.

### *Pre-processing data*

Some of the variables should have been read as a factor variable. According to the census code book, the variable `PwMode` has the following possible values:

- `-3`: Not applicable.
- `1`: Car, truck or van - as a driver.
- `2`: Car, truck or van - as a passenger.
- `3`: Bus.
- `4`: Subway or elevated rail.
- `5`: Light rail, streetcar or commuter train.
- `6`: Passenger ferry.
- `7`: Walked.
- `8`: Bicycle.
- `9`: Motorcycle, scooter or moped.

- 10: Other method.

We'll rename the travel modes to facilitate the readability of the data. Additionally, we'll remove from our analysis travel modes signed as 'Other methods':

```
census <- census %>%
  filter(PwMode < 10) %>%
  mutate(PwMode_label = case_when(
    PwMode == -3 ~ "Not applicable",
    PwMode > 0 & PwMode <= 2 ~ "Car/motor",
    PwMode == 9 ~ "Car/motor",
    PwMode >= 3 & PwMode <= 6 ~ "Transit",
    PwMode == 7 ~ "Walk",
    PwMode == 8 ~ "Bike"),

    PwMode_label = factor(PwMode_label, levels = c("Bike", "Walk", "Car/motor", "Transit", "Not applicable"))
```

Now let's visualize the PwMode label summary:

```
summary(census$PwMode_label)
```

|    |      |      |           |         |                |
|----|------|------|-----------|---------|----------------|
| ## | Bike | Walk | Car/motor | Transit | Not applicable |
| ## | 2673 | 2712 | 61389     | 60304   | 70140          |

A not applicable situation for this variable is when the respondent is not commuting to go to work (unemployed or a person out of the labour force, a person who works from home and therefore does not need to commute).

According to the census documentation, the LBR\_FORC can have the following values:

- -3: Not Applicable, < 15 years
- 1: In Labour Force, Employed
- 2: In Labour Force, Unemployed
- 3: Not in Labour Force

In this case, we can update this variable as bellow:

```
census <- census %>%
  mutate(LBR_FORC_label = case_when(
    LBR_FORC == -3 ~ "Not applicable",
    LBR_FORC == 1 ~ "Employed",
    LBR_FORC == 2 ~ "Unemployed",
    LBR_FORC == 3 ~ "Not in LF"),

    LBR_FORC_label = factor(LBR_FORC_label, levels = c("Employed", "Unemployed", "Not in LF", "Not applicable"))
```

Next chunk update all -3 values to Not applicable for the other categorical variables:

```
census <- census %>%
  mutate(across(c(PWDA, PWPR, PWCD, PWCMA), ~ ifelse(. == -3, "Not applicable", .)))
```

Finishing the pre-processing step, to increase the readability of the table, we will update the name of the provinces of the respondents:

```
census <- census %>%
  mutate(Pr_label = case_when(
    Pr == 10 ~ "Newfoundland and Labrador",
    Pr == 11 ~ "Prince Edward Island",
    Pr == 12 ~ "Nova Scotia",
    Pr == 13 ~ "New Brunswick",
    Pr == 24 ~ "Quebec",
    Pr == 35 ~ "Ontario",
    Pr == 46 ~ "Manitoba",
    Pr == 47 ~ "Saskatchewan",
    Pr == 48 ~ "Alberta",
    Pr == 59 ~ "British Columbia",
    Pr == 60 ~ "Yukon",
    Pr == 61 ~ "Northwest Territories",
    Pr == 62 ~ "Nunavut"),
    Pr_label = as.factor(Pr_label))
```

Visualizing the summary statistics:

```
summary(census)
```

```
##      Frame_ID      PRCDDA      Pr      CMA
## Min.      :    1  Min.    :10010165  Min.    :10.00  Length:197218
## 1st Qu.: 50015  1st Qu.:35061323  1st Qu.:35.00  Class :character
## Median :100002  Median :35204479  Median :35.00  Mode  :character
## Mean    : 99996  Mean    :37273048  Mean    :37.01
## 3rd Qu.:149996  3rd Qu.:47150260  3rd Qu.:47.00
## Max.    :200000  Max.    :62080028  Max.    :62.00
##
##      PCD      CompW1      LBR_FORC      CfInc_AT
## Min.    :1001  Min.    : 1.0  Min.    : -3.000  Min.    : 282
## 1st Qu.:3506  1st Qu.:136.0  1st Qu.: 1.000  1st Qu.: 35674
## Median :3520  Median :271.0  Median : 1.000  Median : 59040
## Mean    :3727  Mean    :270.7  Mean    : 1.154  Mean    : 64688
## 3rd Qu.:4715  3rd Qu.:406.0  3rd Qu.: 2.000  3rd Qu.: 87940
## Max.    :6208  Max.    :540.0  Max.    : 3.000  Max.    :246147
##
```

```
##      CFCNT      CF_PnCF_NumEarnings      PWDA      PWPR
##  Min.   :3.000  Min.   :1.0      Length:197218  Length:197218
##  1st Qu.:3.000  1st Qu.:2.0      Class :character  Class :character
##  Median :4.000  Median :2.0      Mode  :character  Mode  :character
##  Mean   :4.001  Mean   :2.5
##  3rd Qu.:5.000  3rd Qu.:3.0
##  Max.   :5.000  Max.   :4.0
##
##      PWCD      PWCMA      PWDUR      PWDist
##  Length:197218  Length:197218  Min.   : -3.00  Min.   : -3
##  Class :character  Class :character  1st Qu.: -3.00  1st Qu.: -3
##  Mode  :character  Mode  :character  Median : 63.00  Median :22573
##                                     Mean   : 58.36  Mean   :21074
##                                     3rd Qu.:103.00  3rd Qu.:38621
##                                     Max.   :178.00  Max.   :50000
##
##      PwMode      PwMode_label      LBR_FORC_label
##  Min.   : -3.0000  Bike      : 2673  Employed      :127078
##  1st Qu.: -3.0000  Walk      : 2712  Unemployed    : 50037
##  Median : 1.0000  Car/motor :61389  Not in LF     : 10121
##  Mean   : 0.9449  Transit   :60304  Not applicable: 9982
##  3rd Qu.: 3.0000  Not applicable:70140
##  Max.   : 9.0000
##
##      Pr_label
##  Ontario      :89897
##  Quebec        :31074
##  British Columbia:24896
##  Alberta        :19413
##  Saskatchewan   : 7382
##  New Brunswick  : 6644
##  (Other)        :17912
```

Considering that the original census dataset is a very large data file, it can be interesting to filter respondents based on where they live. In this demonstration, we will filter the dataset to include only respondents from the city of Toronto. We can get the information about the city of origin from the PCD variable. The PCD code for the city of Toronto is 3520:

```
census_city <- census %>%
  filter(PCD == 3520)
```



### *Appropriate summary statistics by scale of measurement*

Recall from the previous session that not all operations are defined for all scales of measurement. For example, variables in the nominal scale could be compared using only boolean operators “==” (exactly equal to) and “!=” (not equal to). No arithmetic operations are defined for ordinal data. And division and multiplication are not appropriate for interval data.

This has implications for the kind of statistics that are appropriate by scale of measurement.

Consider a commonly used summary statistic: the mean of a variable. The mean is defined as follows:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

Is it appropriate to calculate the mean of a categorical variable? What is the meaning of two cars plus one bicycle divided by three?

To understand which summary statistics are appropriate, we must know what various summary statistics aim to represent, and how they are calculated.

### *Central tendency*

Summary statistics help in reducing information and provide different perspectives on data. *Central tendency* measures summarize a distribution by identifying a “typical” value, as finding the center of mass of the data. For instance, given a sequence of quantitative values, such as:

```
example_list <- c(20, 30, 32, 34, 41, 41, 45, 46, 48, 51, 53, 54, 54, 56, 57, 58, 58, 59,
  60, 61, 64, 65, 65, 69, 71, 74, 77, 79, 88, 94)
```

This sequence can be visualized in a stem-and-leaf display to understand where the distribution is “heavier,” indicating the center of mass. Central tendency measures help identify this typical value:

| stem | leaf      |
|------|-----------|
| 2    | 0         |
| 3    | 024       |
| 4    | 11568     |
| 5    | 134467889 |
| 6    | 014559    |
| 7    | 1479      |
| 8    | 8         |

| stem | leaf |
|------|------|
| 9    | 4    |

For nominal variables, where categories lack a meaningful order, the concept of central tendency still applies, though the methods differ from those used for ordered or quantitative data.

### Mode

The mode is the most frequent value in a distribution and is useful for both nominal and ordinal variables. To find the mode, we can count the occurrences of each value. We can obtain the mode of the `example_list` by executing the `Mode` function of the `DescTools` library:

```
DescTools::Mode(example_list)

## [1] 41 54 58 65
## attr("freq")
## [1] 2
```

The chunk above shows that the `example_list` contains 4 values as the most frequent, each of them being repeated 2 times: 41, 54, 58, and 65.

Using the `tabyl()` function from the `{janitor}` package, which facilitates pipe-friendly tabulations, we could easily determine the mode of the variable `LBR_FORC` in census table:

```
census_city |>
  tabyl(LBR_FORC_label) # WRONG! You need to consider the weights!

## LBR_FORC_label      n    percent
##      Employed 29888 0.64449907
##      Unemployed 11769 0.25378445
##      Not in LF  2351 0.05069651
## Not applicable  2366 0.05101997
```

The output shows that the `Employed` category is the mode of the `LBR_FORC` variable (considering the test data), representing around 75% of the cases. *However*, it is important to remember that the census survey contains the weight of each respondent, described by the variable `Compw1`. This means that all the statistics made for the general population need to consider that person's weight. To do this, we will use the `wtd.mode` function from the R package `poliscidata`:

```
poliscidata::wtd.mode(census_city$LBR_FORC_label, weights = census_city$CompW1)

## [1] "Employed"
```

### Median

The median is the quantile that splits a quantitative variables in two parts of equal size, the bottom 50% and the top 50% of values.

To check the median value of total income of the family considering the weights, we can use the `Hmisc` library:

```
census_city %>%
  dplyr::summarise(Median = Hmisc::wtd.quantile(CfInc_AT, weights = CompW1, probs = 0.5))

## # A tibble: 1 x 1
##   Median
##   <dbl>
## 1  59020
```

This value is slightly different (but needs to be considered!) than the regular median without considering the weights:

```
median(census_city$CfInc_AT) # WRONG! You need to consider the weights!

## [1] 59043.36
```

### Mean

The mean is probably the best known measure of central tendency, and it is defined as the sum of the values divided by the number of observations. Since it involves arithmetic operations it is not appropriate for categorical variables.

To check the average value of a regular data set (without weights), we can use the `mean()` built-in function. In our case, to consider the weights of the respondents, we can again apply the `Hmisc` library:

```
#mean(census$CfInc_AT) # WRONG! You need to consider the weights!

census_city %>%
  dplyr::summarise(Mean = Hmisc::wtd.mean(CfInc_AT, weights = CompW1))

## # A tibble: 1 x 1
##   Mean
##   <dbl>
## 1 64595.
```

Another way of considering the weights to obtain the mean is to apply the weighted average. The mean value in both methods is the same:

```
census_city %>%
  dplyr::summarise(Mean = sum(CfInc_AT * CompW1)/sum(CompW1))
```

```
## # A tibble: 1 x 1
##   Mean
##   <dbl>
## 1 64595.
```

### *Spread*

Another important property of a distribution of values is how wide or compact it is. Compare the two stem-and-leaf tables below.

| stem | leaf      |
|------|-----------|
| 2    | 0         |
| 3    | 024       |
| 4    | 11568     |
| 5    | 134467889 |
| 6    | 014559    |
| 7    | 1479      |
| 8    | 8         |
| 9    | 4         |

| stem | leaf  |
|------|-------|
| 1    | 48    |
| 2    | 08    |
| 3    | 024   |
| 4    | 1156  |
| 5    | 13789 |
| 6    | 01459 |
| 7    | 149   |
| 8    | 468   |
| 9    | 45    |
| 10   | 7     |

The first stem-and-leaf table is more “compact”: the tails of the distribution are closer together and the center of mass is “heavier”, compared to the second table, that has a wider spread.

### *Minimum and maximum*

The minimum and maximum values give an idea of how spread the distribution is. In the first of the preceding tables the minimum is 20 and the maximum is 94. In the second table, the minimum is 14 and the maximum is 107. The *range* is the difference between the maximum and the minimum:

```
94 - 20
```

```
## [1] 74
```

```
107 - 14
```

```
## [1] 93
```

The second distribution is more spread. Both the minimum and maximum can be obtained without considering the weights of the population.

### *Inter-quartile range*

The inter-quartile range is similar to the range, but instead of being calculated using the minimum and maximum values of the distribution, it uses the third and first quartiles. Quartiles are a form of quantile that divides a sequence of values in four equal parts, so the second quantile represents the value that separates the lowest 25% of the sample from the remaining 75%, and the third quantile is the value that splits the highest 25% of the sample from the lowest 75%.

Previously, we used the `Hmisc` library to obtain the median value of `CfInc_AT`. In fact, we applied the quantile function to obtain the median value considering the weights, because the median is the value that divides the values into two quantiles, those below and those above the median value. Now, we will obtain the values for the second and third quantiles by running the chunk below:

```
census_city %>%
  dplyr::summarise(Second_Q = Hmisc::wtd.quantile(CfInc_AT, weights = CompW1, probs = 0.25),
                  Median = Hmisc::wtd.quantile(CfInc_AT, weights = CompW1, probs = 0.5),
                  Third_Q = Hmisc::wtd.quantile(CfInc_AT, weights = CompW1, probs = 0.75))

## # A tibble: 1 x 3
##   Second_Q Median Third_Q
##   <dbl>   <dbl>   <dbl>
## 1   35472.  59020   88263
```

To obtain the inter-quartile range (IQR) we just need to subtract the second quantile from the third quantile:

```
census_city %>%
  dplyr::summarise(Second_Q = Hmisc::wtd.quantile(CfInc_AT, weights = CompW1, probs = 0.25),
                  Median = Hmisc::wtd.quantile(CfInc_AT, weights = CompW1, probs = 0.5),
                  Third_Q = Hmisc::wtd.quantile(CfInc_AT, weights = CompW1, probs = 0.75),
                  IQR = Third_Q - Second_Q)
```

```
## # A tibble: 1 x 4
##   Second_Q Median Third_Q   IQR
##   <dbl>   <dbl>   <dbl> <dbl>
## 1   35472.  59020   88263 52791.
```

The inter-quartile range involves an arithmetic operation, which is why it is not an appropriate statistic for categorical variables.

### *Variance and standard deviation*

The variance is another widely used measure of the spread of a distribution. It is defined as:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

In this formula,  $\bar{x}$  is the mean of  $x$  and  $n$  is the number of observations in the sample. Accordingly,  $x_i - \bar{x}$  is the deviation of  $x_i$  from the mean of  $x$ . If we rewrite this as follows:

$$z_i = (x_i - \bar{x})^2$$

It is easy to see that the variance is actually the mean of the square of the deviations from the mean:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n z_i$$

The standard deviation is simply the square root of the variance and returns the variance to the same units as the original variable. To consider the weights in the calculation of the variance and standard deviation, you can execute the chunk bellow:

```
census_city %>%
  dplyr::summarise(Mean = Hmisc::wtd.mean(CfInc_AT, weights = CompW1),
                  Variance = Hmisc::wtd.var(CfInc_AT, weights = CompW1),
                  SD = sqrt(Variance))

## # A tibble: 1 x 3
##   Mean      Variance      SD
##   <dbl>      <dbl>   <dbl>
## 1  64595. 1396215890. 37366.
```

### *Univariate description*

Summary statistics of central tendency and spread refer to a single variable and are appropriately called univariate descriptors. These descriptors are very important, and we neglect exploring them at our

own peril. They often tell us important aspects of the data, including how complete a data set is, how much variation is there, whether there are atypical or unusual values.

As an example, let us calculate the mean, standard deviation, and maximum of CfInc\_AT (total income of census family):

```
CfInc_AT_statistics <- census_city %>%
  dplyr::summarise(Mean = Hmisc::wtd.mean(CfInc_AT, weights = CompW1),
                   Median = Hmisc::wtd.quantile(CfInc_AT, weights = CompW1, probs = 0.5),
                   SD = sqrt(Hmisc::wtd.var(CfInc_AT, weights = CompW1)),
                   Max = max(CfInc_AT),
                   Min = min(CfInc_AT))
```

The maximum average income of the person responsible for the household in the test data set for city of Toronto was R\$ 259,872.70. Just how common or unusual is this value? That depends on how close (or far away) from the mean of the distribution this is, as well as on the spread of the distribution. The deviation from the mean is:

```
CfInc_AT_statistics$Max - CfInc_AT_statistics$Mean
```

```
## [1] 181551.8
```

That is, approximately R\$ 159,762. But the typical deviation from the mean in the sample was about R\$ 62,009! Now, calculating:

```
(CfInc_AT_statistics$Max - CfInc_AT_statistics$Mean)/CfInc_AT_statistics$SD
```

```
## [1] 4.858747
```

This tells us that the census tract with the highest total family income receives almost three times more than the average total family income. This observation is indeed quite unusual.

How unusual was the census tract with the lowest average income? Let us retrieve the minimum duration:

```
CfInc_AT_statistics$Min
```

```
## [1] 658.96
```

That is, around \$ 3001. Again, the typical deviation from the mean in the sample was R\$ 62,009! So, calculating:

```
(CfInc_AT_statistics$Min - CfInc_AT_statistics$Mean)/CfInc_AT_statistics$SD
```

```
## [1] -1.711077
```

The census tract with the lowest average income is closer to the mean, and approximately one and half standard deviation below the mean. Univariate description is a powerful way to get to know our data before doing any more sophisticated explorations or analysis.

## Bivariate description

Moving on from univariate description, understanding how two variables relate to one another is another key aspect of EDA.

### Categorical variables: cross-tabulations

Univariate description of a categorical variable involves tabulating the number of instances of each response. This can be expanded to simultaneously tabulating two categorical variables. We will summarise the variables `Pr` (Province) and `PwMode` (mode of transport), summing all weights for each combination:

```
census %>%
  group_by(Pr, PwMode) %>%
  summarise(weighted_n = sum(CompW1, na.rm = TRUE), .groups = "drop")

## # A tibble: 130 x 3
##       Pr PwMode weighted_n
##   <dbl> <dbl>     <dbl>
## 1     10     -3    230190
## 2     10      1    169531
## 3     10      2      8004
## 4     10      3   132954
## 5     10      4    31733
## 6     10      5    14509
## 7     10      6    28408
## 8     10      7    10834
## 9     10      8     5991
## 10    10      9    32887
## # i 120 more rows
```

To facilitate the visualization, we will apply a pivot technique, turning the long table into a wide one like a contingency table:

```
census_city %>%
  group_by(Pr_label, PwMode_label) %>%
  summarise(weighted_n = sum(CompW1, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = PwMode_label, values_from = weighted_n, values_fill = 0)

## # A tibble: 1 x 6
##   Pr_label   Bike   Walk `Car/motor` Transit `Not applicable`
##   <fct>     <dbl> <dbl>     <dbl>   <dbl>         <dbl>
## 1 Ontario  199208 202961   3905538 3823275       4458300
```

What do we learn from this table? The output of the previous chunk can be *adorned*, which can improve the readability of the table.



The following table gives the total sum of the columns as a row at the bottom of the table:

```
census_city %>%
  group_by(Pr_label, PwMode_label) %>%
  summarise(weighted_n = sum(CompW1, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = PwMode_label, values_from = weighted_n, values_fill = 0) %>%
  adorn_totals("row") %>%
  adorn_totals("col") %>%
  adorn_percentages("row") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

| ## | Pr_label          | Bike                | Walk           | Car/motor         | Transit           |
|----|-------------------|---------------------|----------------|-------------------|-------------------|
| ## | Ontario           | 1.6% (199,208)      | 1.6% (202,961) | 31.0% (3,905,538) | 30.4% (3,823,275) |
| ## | Total             | 1.6% (199,208)      | 1.6% (202,961) | 31.0% (3,905,538) | 30.4% (3,823,275) |
| ## | Not applicable    |                     | Total          |                   |                   |
| ## | 35.4% (4,458,300) | 100.0% (12,589,282) |                |                   |                   |
| ## | 35.4% (4,458,300) | 100.0% (12,589,282) |                |                   |                   |

### Quantitative variables: correlation

The mean and standard deviation are key univariate descriptors of quantitative variables. When we are interested in the relationship between two quantitative variables we use a related concept, the *covariance*. The covariance is the mean of the product of the deviations from the mean of two variables:

$$C(x, y) = \frac{1}{n} \sum_i (x_i - \bar{x})(y_i - \bar{y})$$

Here,  $\bar{x}$  and  $\bar{y}$  are the means of the variables. Positive or negative deviations result in positive covariance, while opposing deviations result in negative covariance. Like-like deviations (both positive or negative) increase covariance, while opposite deviations decrease it. Covariance can be normalized by dividing it by the product of the variables' standard deviations, resulting in the correlation coefficient:

$$r(x, y) = \frac{C(x, y)}{\sigma_x \cdot \sigma_y}$$

The correlation coefficient ranges between -1 and 1, where 0 indicates no covariance. Correlations can be calculated using the `wtd.cor` function from the `{weights}` package in a pipe-friendly manner:

```
# PWDUR: Commuting duration
# PWDist: Distance (straight-line) from home to work

weights::wtd.cor(x = census_city$PWDUR, y = census_city$PWDist, weight = census_city$CompW1)
```

```
## correlation      std.err  t.value p.value
## Y    0.9475334  0.001484417 638.3201      0
```

The code above shows that the variables `PWDUR` and `PWDist` have a very high correlation (above 0.95), which makes sense, as travel time tends to be longer when the distance from the destination is high.

### *Visualization techniques*

Summary statistics are data reduction techniques that focus on specific characteristics of the data, such as central tendency or spread. These statistics provide a single number to describe these characteristics but are inherently limited as they do not capture all aspects of the data. This limitation is intentional, as the goal of Exploratory Data Analysis (EDA) is to help us understand data without overwhelming our cognitive capabilities.

While summary statistics are valuable, visualization techniques serve as a complementary approach. Statistical plots leverage the brain's exceptional ability to process visual information, overcoming the limitations of short-term memory retention when handling alphanumeric data.

What is the central tendency of `CfInc_AT` (total income of census family ) taking into account only the numbers shown? Is the `PWDist` more or less spread than the `CfInc_AT` (distance (straight-line) from home to work)? These properties of the data are not readily evident from a quick visual scan of the numbers. Summary statistics retrieve the desired information for us by “flattening” the data:

```
census |>
  filter(PWDist != -3) |>
  dplyr::select(CfInc_AT, PWDist) |>
  slice_head(n = 10) |>
  summary()
```

```
##      CfInc_AT      PWDist
## Min.   : 21960   Min.    : 5187
## 1st Qu.: 56934   1st Qu.:14363
## Median : 80208   Median :19020
## Mean   : 78200   Mean    :23989
## 3rd Qu.: 91077   3rd Qu.:35998
## Max.   :175716   Max.    :47486
```

To make matters more difficult, this is only a small part of the full table (only ten rows and six columns). Additionally, the summary does not account the weights of the respondents. The task of iden-

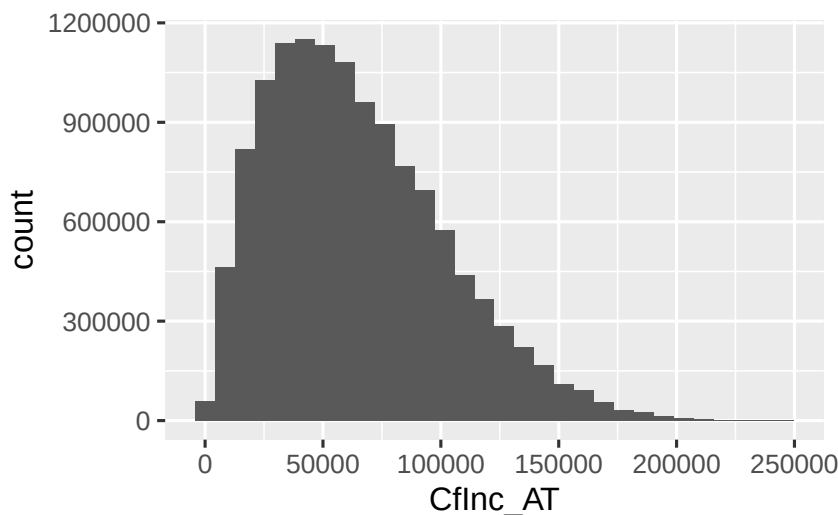
tifying patterns becomes increasingly complicated as the number of observations and the number of variables grow.

The grammar of graphics, formalized by Leland Wilkinson in his book, provides a structured approach to creating statistical plots, similar to how the grammar of data manipulation offers a way to think about data operations. This concept inspired the creation of the `{ggplot2}` package, which allows for flexible and intuitive plot creation by layering graphical elements. While `{ggplot2}` requires explicit specification of what and how to plot, making it more verbose than simple plotting functions, it enables the creation of more sophisticated and expressive plots. This approach integrates data manipulation and plotting, allowing for detailed and customized visualizations. As with summary statistics, it's crucial to consider the scale of measurement when selecting geometric objects for plots, which vary based on whether the plots are univariate, bivariate, or multivariate, and if the variables are categorical or quantitative.

### *Univariate plots*

Univariate description involves exploring the main attributes of a single variable, typically its central tendency and spread. For quantitative variables, an appropriate geometric object is a histogram, implemented as `geom_hist()`. Using the variable `CfInc_AT` (total income of census family), we have:

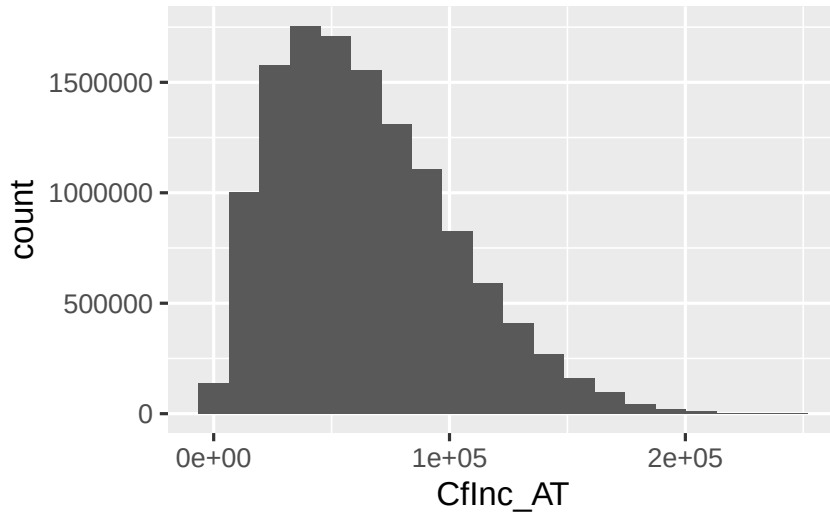
```
ggplot(data = census_city,
       aes(x = CfInc_AT, weight = CompW1)) +
  geom_histogram()
```



A histogram is the number of cases (the *count* of cases) by ranges of values. We only need to encode a single variable (in the example

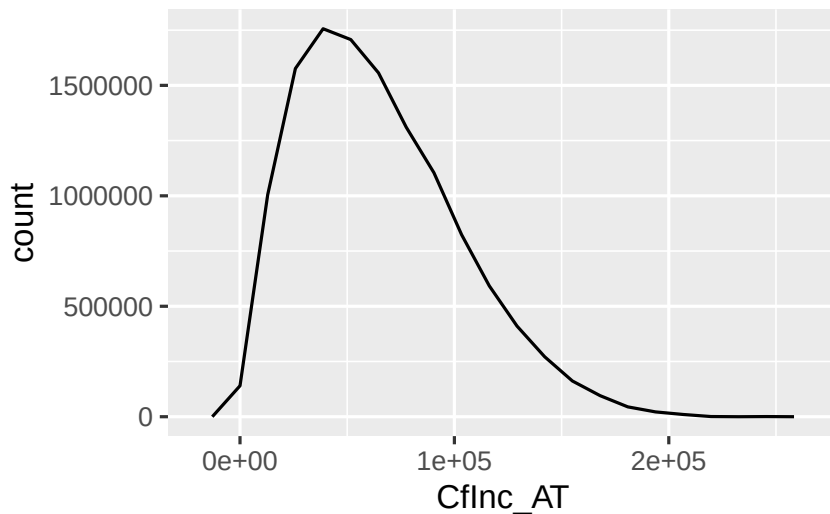
above the `CfInc_AT`), because the “count” on the y-axis is a computed statistic. The default number of bins in `geom_hist()` is 30, but this can be adjusted:

```
ggplot(data = census_city,
       aes(x = CfInc_AT, weight = CompW1)) +
  geom_histogram(bins = 20)
```



An alternative geometric object is a frequency polygon, as shown here:

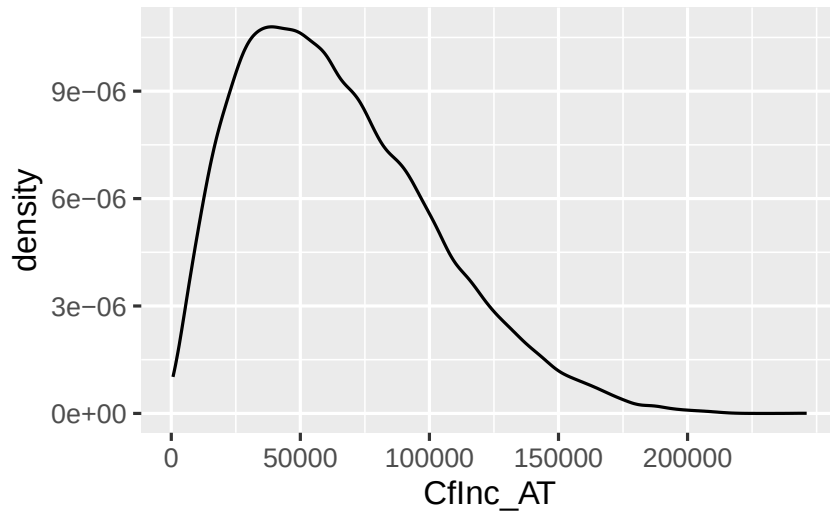
```
ggplot(data = census_city,
       aes(x = CfInc_AT, weight = CompW1)) +
  geom_freqpoly(bins = 20)
```



A density plot is a smoother version of a frequency polygon:

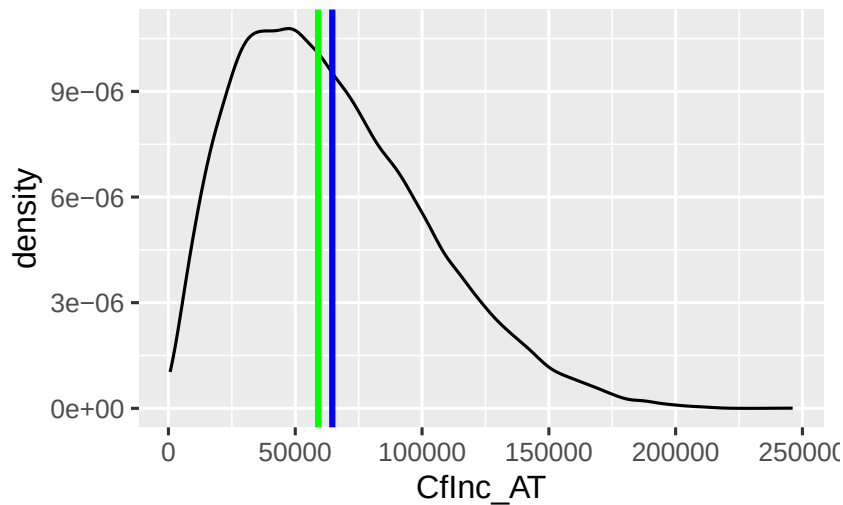
```
ggplot(data = census_city,
       aes(x = CfInc_AT, weight = CompW1)) +
```

```
geom_density(bins = 20) +
scale_x_continuous(breaks=seq(0,max(census_city$CfInc_AT),50000))
```



We can use `geom_vline()` to draw vertical lines in the plot (the mean in blue, the median in green):

```
ggplot(data = census_city,
       aes(x = CfInc_AT)) +
geom_density(bins = 20) +
geom_vline(xintercept = CfInc_AT_statistics$Mean,
           color = "blue",
           size = 1) +
geom_vline(xintercept = CfInc_AT_statistics$Median,
           color = "green",
           size = 1)
```

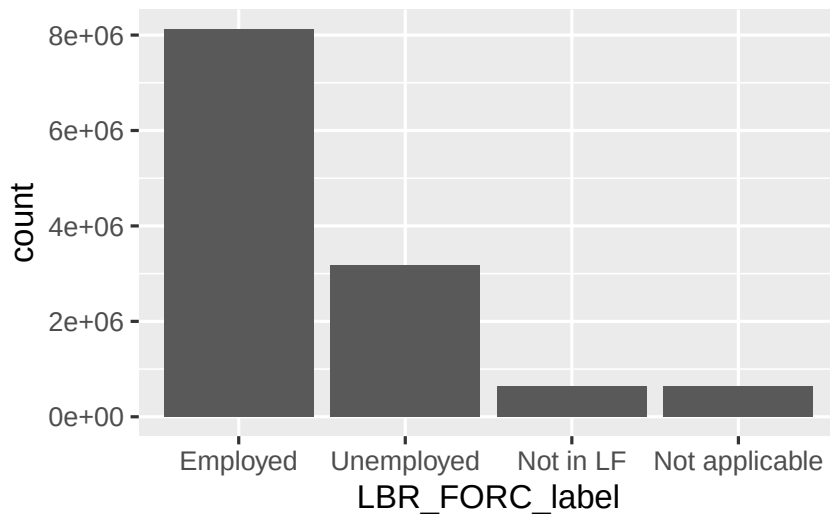


The mean and median of a distribution can differ due to the distribution's lack of symmetry. The mean is influenced by extreme values

or outliers, making it less robust compared to the median, which is more stable as it is not affected by unusual values.

For categorical variables, a bar chart, implemented with `geom_bar()`, is the appropriate geometric object. While bar charts may resemble histograms, they differ in two key ways: the order of categories is not essential, and there are no value ranges, only distinct category labels.

```
ggplot() +  
  geom_bar(data = census_city,  
           aes(x = LBR_FORC_label, weight = CompW1))
```

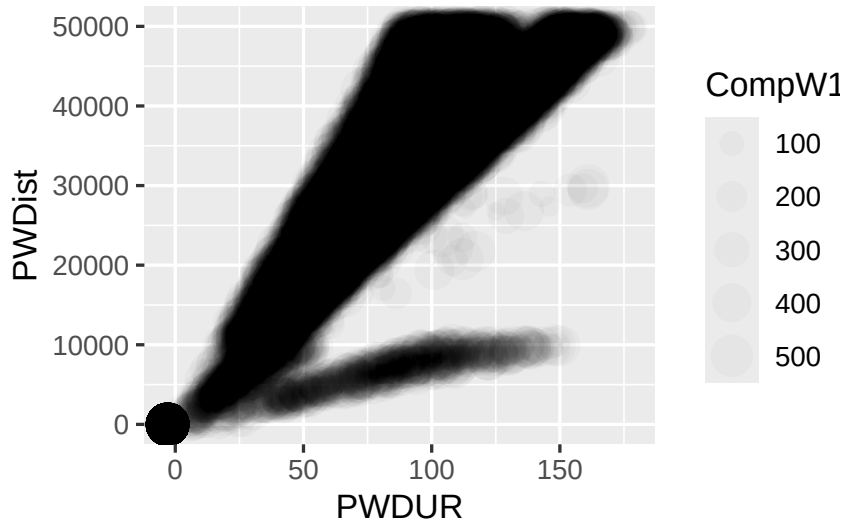


### *Bivariate plots*

- Two quantitative variables

The scatterplot is a fundamental visualization method for exploring the relationship between two quantitative variables. It maps the values of these variables as points on the x- and y-axes, allowing for the examination of patterns, correlations, and trends between them. To account with the weights, we will size the points based on the weight of the respondents.

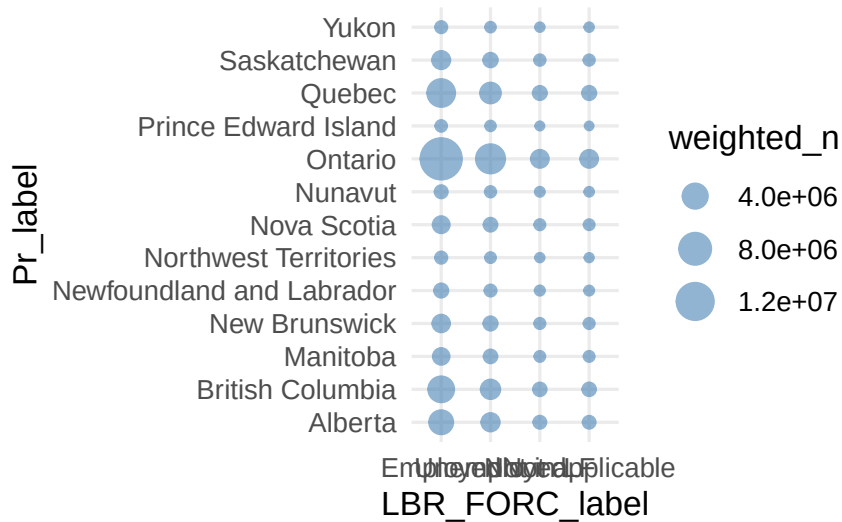
```
ggplot(data = census_city) +  
  geom_point(aes(x = PWDUR, y = PWDist, size = CompW1),  
            alpha = 0.025)
```



### Two categorical variables

Two categorical variables can be explored by means of count plots:

```
census %>%
  group_by(LBR_FORC_label, Pr_label) %>%
  summarise(weighted_n = sum(CompW1, na.rm = TRUE), .groups = "drop") %>%
  ggplot(aes(x = LBR_FORC_label, y = Pr_label)) +
  geom_point(aes(size = weighted_n), color = "steelblue", alpha = 0.6) +
  theme_minimal()
```



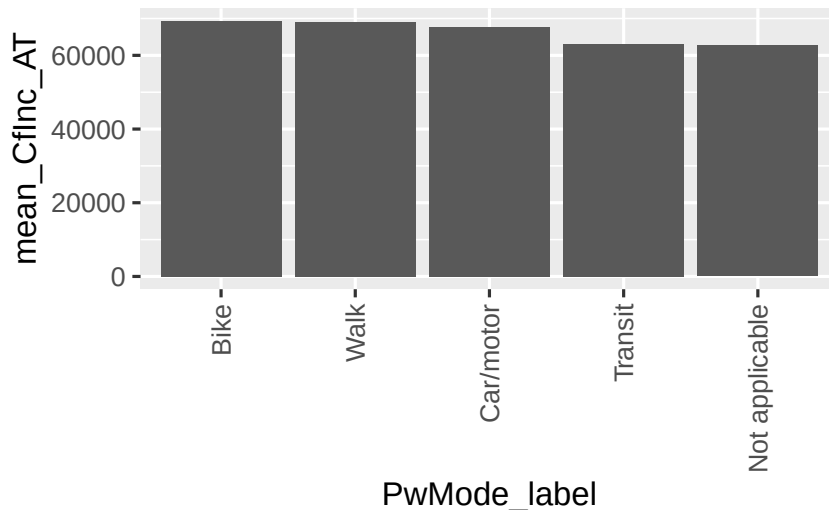
Count plots are a visual alternative to a cross-tabulation.

### One categorical and one quantitative variable

Visualization techniques can accommodate combinations of one categorical and one quantitative variable, allowing for more nuanced

exploration of data. For instance, column plots can map a categorical variable to one axis and a quantitative variable to the other. In this approach, summary statistics like the mean of a quantitative variable (e.g., `CfInc_AT`) are calculated for each category of a categorical variable (e.g., commute mode) and visualized using `geom_col()`. This technique helps to clearly display differences and trends across categories.

```
census_city |>
  group_by(PwMode_label) |>
  dplyr::summarize(mean_CfInc_AT = Hmisc::wtd.mean(CfInc_AT, weights = CompW1)) %>%
  ggplot() +
  geom_col(aes(x = PwMode_label,
               y = mean_CfInc_AT)) +
  scale_x_discrete(guide = guide_axis(angle = 90))
```



The graph shows that respondents who commute by bicycle have higher household incomes, while people who don't commute by bicycle tend to have lower household incomes. The boxplot is another valuable visualization technique. It uses a rectangular box to represent the interquartile range (IQR) of a distribution, with lines (whiskers) extending to 1.5 times the IQR to show the range of most data points. Extreme values beyond this range are depicted as individual dots. The median of the distribution is shown as a line within the box. In `ggplot2`, boxplots are created using `geom_boxplot()`.

First, to consider the weights in the boxplot, we will expand repeat each line of the census dataset according to their respective weights:

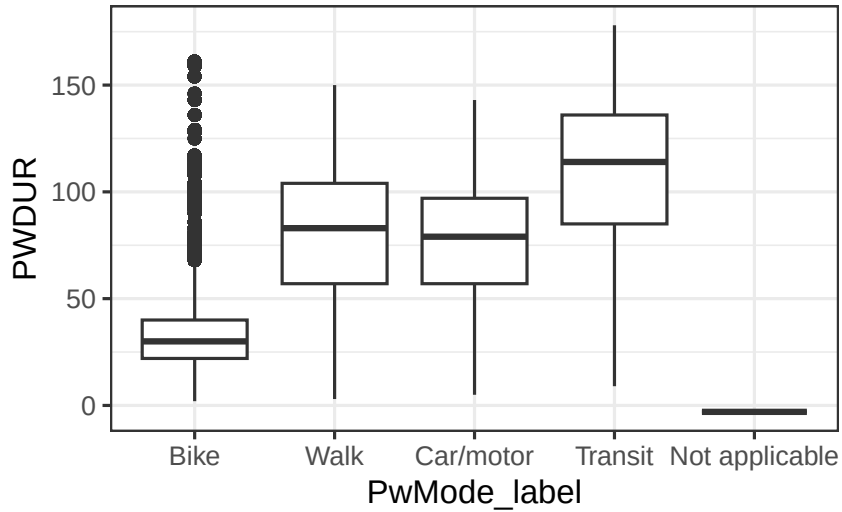
```
census_city_expanded <- census_city %>%
  uncount(weights = round(CompW1))
```

Now, we can visualize the boxplot of the duration by the type of



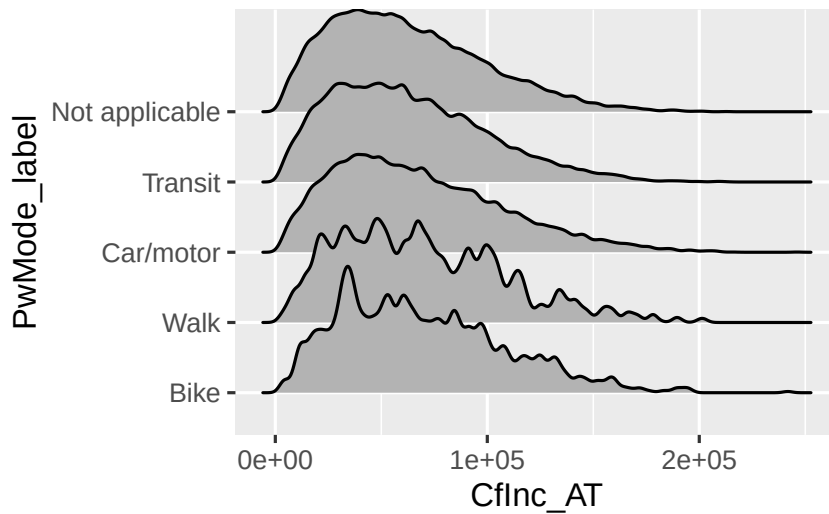
transportation mode:

```
census_city_expanded %>%
  ggplot(aes(x = PwMode_label, y = PWDUR)) +
  geom_boxplot() +
  theme_bw()
```



The boxplot does obscure some of the detail of the underlying distribution of values. Ridge plots address this by plotting the density of the distribution instead:

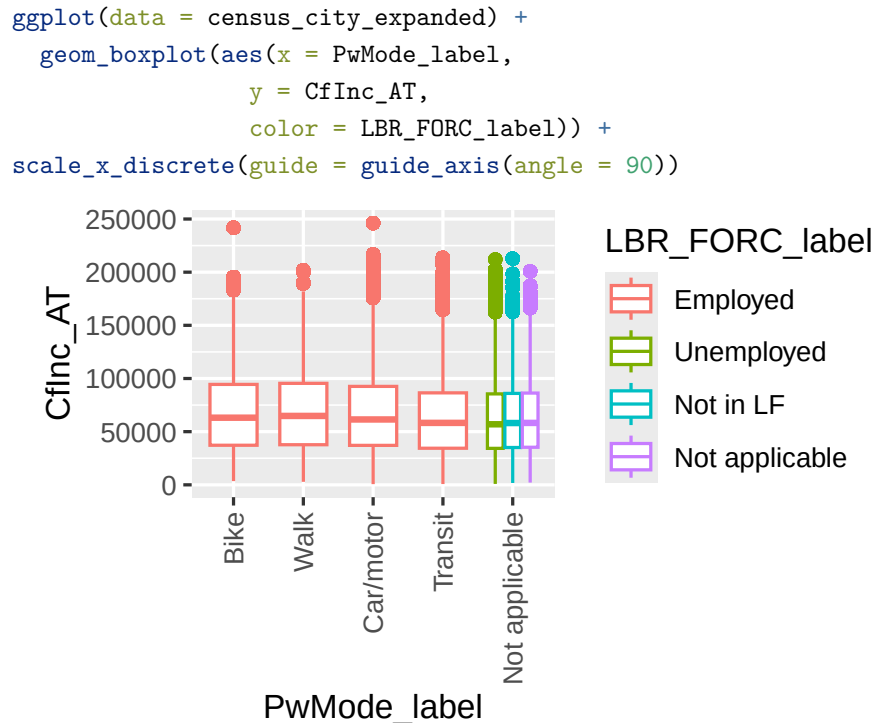
```
ggplot(data = census_city_expanded) +
  geom_density_ridges(aes(x = CfInc_AT, y = PwMode_label))
```



### *Multivariate description*

Higher dimensional visualization can be created by encoding additional variables using available aesthetics. The following chunk of

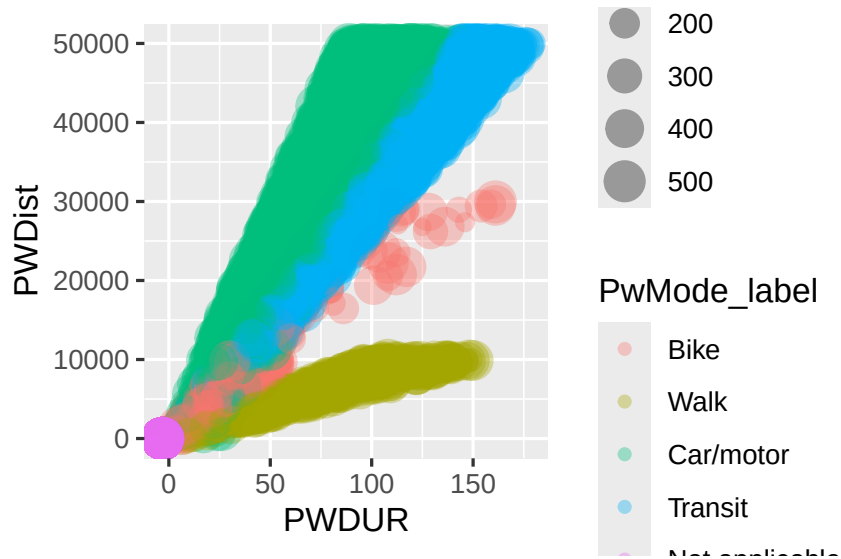
code recreates the boxplot of `PwMode` and `CfInc_AT`, and further maps `LBR_FORC` to color:



This graph shows that the household income of people who commute using active modes tends to be higher. As you need to be employed to commute, the difference variable `LBR_FORC` only appears for those who are not commuting to work (`not applicable`) and shows that unemployed people have the lowest household income.

The following example recreates the scatterplot of `PWDUR` and `PWDist` that we did before, but now adds `PwMode` to plot, encoded to color and shape:

```
ggplot(data = census_city) +
  geom_point(aes(x = PWDUR,
                y = PWDist,
                color = PwMode_label,
                size = CompW1),
            alpha = 0.35)
```



The linear relation displayed in the scatter plot between the variables PWDUR and PwMode occurs because this data was generated synthetically.